THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE MCQ TEST SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The solution of $2x + y = 10$, $x - y = 2$ has $x =$ 1

(a) 4 **(b)** -4 **(c)** 6 **(d)** 2

2. For $x^2 - 8x + k = 0$ to have equal roots, $k =$ 1

(a) 4 **(b)** 16 **(c)** 64 **(d)** 8

3. The 8th term from the end of the AP with last term 80 and $d = 3$ is: 1

(a) 59 **(b)** 62 **(c)** 56 **(d)** 101

4. $\triangle ABC \sim \triangle DEF$ with areas 18 cm² and 32 cm². If $AB = 9$ cm, then DE (cm) $=$ 1

(a) 48 **(b)** 15 **(c)** 9 **(d)** 12

5. The point which divides the join of $(3, 3)$ and $(9, 12)$ in the ratio 1:2 internally is: 1

(a) $(5,\ 6)$ **(b)** $(7,\ 9)$ **(c)** $(6,\ \frac{15}{2})$ **(d)** $(9,\ 12)$

6. If $\tan A = \frac{12}{5}$, then $\sin A =$ 1

(a) $\frac{12}{5}$ **(b)** $\frac{12}{13}$ **(c)** $\frac{13}{12}$ **(d)** $\frac{5}{13}$

7. From a point on level ground, 21 m away from the foot of a vertical tower, the angle of elevation of its top is 30°. The height of the tower is: 1

(a) 7 m **(b)** $21\sqrt{3}$ m **(c)** $7\sqrt{3}$ m **(d)** 21 m

8. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that $OQ = 12$ cm. The length of PQ is: 1

(a) 8.5 cm **(b)** 12 cm **(c)** 13 cm **(d)** $\sqrt{119}$ cm

9. A circular field has radius 21 m. Fencing costs ₹ 5 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 6930 **(b)** ₹ 660 **(c)** ₹ 330 **(d)** ₹ 132

10. A toy is a cone of base radius 21 cm and height 28 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 4620 cm² **(b)** 2310 cm² **(c)** 5082 cm² **(d)** 6468 cm²

11. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 2, 17, 12, 10, 9 respectively, The median of the data is: 1

(a) 70 **(b)** 60 **(c)** 66.4 **(d)** 65

12. A coin is tossed three times. The probability of getting at least two heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{4}{9}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{3}{8}$

13. If HCF(510, 92) = 2, then LCM(510, 92) is: 1

(a) 602 **(b)** 23460 **(c)** 46920 **(d)** 1020

14. If one zero of $x^2 - 9x + 18$ is 3, the other zero is: 1

(a) -6 **(b)** 12 **(c)** 18 **(d)** 6

15. The solution of $3x + 2y = 14$, $2x - 3y = 5$ has $y =$ 1

(a) -1 **(b)** 5 **(c)** 1 **(d)** 4

16. If one root of $x^2 - 11x + k = 0$ is 3, then k is: 1

(a) 27 **(b)** 8 **(c)** 11 **(d)** 24

17. If $a_n = 3n + 5$, then the common difference is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 8

18. The length of the diagonal of a square of side 5 cm, squared, is: 1

(a) 100 **(b)** 50 **(c)** 25 **(d)** 10

19. The area of the triangle with vertices $(0, 0)$, $(8, 0)$ and $(0, 5)$ is: 1

(a) 40 **(b)** 13 **(c)** 20 **(d)** $\frac{13}{2}$

20. If $\cos A = \frac{8}{17}$, then $\tan A =$ 1

(a) $\frac{15}{17}$ **(b)** $\frac{15}{8}$ **(c)** $\frac{8}{15}$ **(d)** $\frac{17}{8}$

21. A ladder 16 m long leans against a vertical wall and makes an angle of 60° with the level ground. The height reached on the wall is: 1

(a) 8 m **(b)** $8\sqrt{3}$ m **(c)** $16\sqrt{3}$ m **(d)** $4\sqrt{3}$ m

22. A point P is 35 cm from the centre O of a circle of radius 21 cm. The length of the tangent drawn from P to the circle is: 1

(a) 28 cm **(b)** 56 cm **(c)** 14 cm **(d)** 784 cm

23. A wheel has radius 35 cm. The number of complete revolutions it makes to cover 110 m is: 1

(a) 25 **(b)** 60 **(c)** 100 **(d)** 50

24. A capsule is a cylinder of radius 7 cm and length 14 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 924 cm² **(b)** 2464 cm² **(c)** 1232 cm² **(d)** 1540 cm²

25. For the data with classes 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 with frequencies 3, 13, 5, 16, 7, 6 respectively, The mode of the data is: 1

(a) 80 **(b)** 80.8 **(c)** 90 **(d)** 85.5

26. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is the letter B is: 1

(a) $\frac{9}{11}$ **(b)** $\frac{3}{11}$ **(c)** $\frac{2}{11}$ **(d)** $\frac{1}{11}$

27. The product of a non-zero rational and an irrational number such as $3\sqrt{5}$ is: 1

(a) always irrational **(b)** always rational

(c) always an integer **(d)** sometimes rational

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28. A quadratic polynomial whose zeroes are -2 and 5 is: 1

(a) $x^2 + 3x + 10$ **(b)** $x^2 - 3x + 10$ **(c)** $x^2 + 3x - 10$ **(d)** $x^2 - 3x - 10$

29. For the pair $x + y = 14$ and $x - y = 4$, the value of xy is: 1

(a) 14 **(b)** 53 **(c)** 90 **(d)** 45

30. The roots of $x^2 - 2x - 24 = 0$ are: 1

(a) 4 and 6 **(b)** 4 and -6 **(c)** -4 and -6 **(d)** -4 and 6

31. If $S_n = n^2 + 3n$, then $a_1 =$ 1

(a) 5 **(b)** 4 **(c)** 1 **(d)** 3

32. A man goes 24 m due east and then 32 m due north. His distance from the start (m) is: 1

(a) 1600 **(b)** 56 **(c)** 8 **(d)** 40

33. If the midpoint of $(4, 1)$ and (x, y) is $(7, 2)$, then $x =$ 1

(a) $\frac{11}{2}$ **(b)** 10 **(c)** 3 **(d)** 18

34. The value of $\tan^2 60^\circ - \tan^2 30^\circ$ is: 1

(a) 3 **(b)** $\frac{16}{3}$ **(c)** $\frac{8}{3}$ **(d)** $\frac{11}{3}$

35. A boy whose eyes are 3 m above the ground stands 15 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) $18\sqrt{3}$ m **(b)** 18 m **(c)** 15 m **(d)** 19 m

36. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 12$ cm, $BC = 10$ cm and $CD = 8$ cm. The length of AD is: 1

(a) 30 cm **(b)** 10 cm **(c)** 14 cm **(d)** 6 cm

37. The radii of two circles are 8 cm and 6 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 14 cm **(b)** 28 cm **(c)** 10 cm **(d)** 7 cm

38. A solid is a cube of edge 28 cm surmounted by a hemisphere of diameter 28 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 5320 cm² **(b)** 4704 cm² **(c)** 4088 cm² **(d)** 5936 cm²

39. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 5, 4, 10, 4, 2 respectively, The modal class is: 1

(a) 10–20 **(b)** 30–40 **(c)** 20–30 **(d)** 40–50

40. A bag contains 5 red and 5 blue balls. 5 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{3}{5}$ **(b)** $\frac{11}{15}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

41. The decimal expansion of $\frac{17}{8}$ terminates after how many places of decimals? 1

(a) 5 **(b)** 3 **(c)** 6 **(d)** 4

42. If α, β are zeroes of $x^2 - 5x - 24$, then $\alpha + \beta$ is: 1

(a) -5 **(b)** 5 **(c)** -24 **(d)** 6

43. The pair $x + 3y = 4$ and $3x + ky = 13$ has no solution when $k =$ 1

(a) 10 **(b)** 9 **(c)** 12 **(d)** 3

44. The discriminant of $3x^2 + 2x + 5 = 0$ is: 1

(a) 64 **(b)** -56 **(c)** 56 **(d)** -26

45. The sum of first 8 positive odd numbers is: 1

(a) 72 **(b)** 128 **(c)** 64 **(d)** 56

46. In $\triangle ABC$, $AB = 7$, $BC = 8$, $AC = 10$. The triangle is: 1

(a) equilateral **(b)** not right-angled **(c)** right-angled **(d)** isosceles right

47. The distance of $(-7, 2)$ from the y-axis is: 1

(a) 2 **(b)** 9 **(c)** -7 **(d)** 7

48. $\tan 35^\circ \cdot \tan 55^\circ =$ 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 35

49. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 30 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) $15\sqrt{3}$ m **(b)** $30\sqrt{3}$ m **(c)** $10\sqrt{3}$ m **(d)** 15 m

50. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 70^\circ$. The $\angle OAB$ is: 1

(a) 20° **(b)** 35° **(c)** 70° **(d)** 55°

END OF QUESTION PAPER **Total: 50 Marks**

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